ORION Paper III: A Recoverable Global Knowledge Portrait from Source-Local Scientific Projections

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Abstract

Cross-domain scientific synthesis is vulnerable to false integration: identical names may denote different referents, nominally similar constructs may use incompatible operationalizations, and apparently contradictory claims may differ in context, modality or attribution. We present a recoverable global knowledge portrait built from source-local scientific projections. Before global integration, proposed mappings distinguish referent identity, construct identity, measurement/operationalization, context and representation equivalence. Compatible projections may be GLUEd under explicit preservation conditions; incompatible or unresolved views remain typed OBSTRUCTION or PLURAL_VIEW charts rather than being forced into one canonical summary. The resulting portrait preserves enough lineage to recover its contributing source views and mapping assumptions. Local exact worlds falsify our semantics, but real cross-domain construct validity remains `CANNOT_CHECK` until the external gold study is executed under the bound protocol. We compare against six baselines (long-context, RAG, cross-domain RAG, universal schema, SCOPE/SCION-like fusion, and provenance-aware schema contracts) and eight ablations across four disciplines (biomedicine, physics, social science, climate science).

Keywords: scientific knowledge integration; provenance; schema alignment; measurement harmonization; scientific pluralism; knowledge representation.

1 Introduction

The global scientific literature is increasingly fragmented across disciplines, each with its own terminology, measurement practices, and epistemic norms. Integrating claims from diverse sources into a coherent body of knowledge is a central challenge for scientific knowledge representation, literature-based discovery, and cross-domain synthesis.

Consider a simple example: two papers both mention “zebrafish.” One uses it as a model organism in developmental biology; the other studies it as an invasive species in ecology. The surface term is the same, but the referent is different — a scientific knowledge integrator that treats them as the same entity silently produces a false merge. Conversely, two papers in different subfields may use different terms (“evaluative conditioning” and “affective priming”) to refer to the same underlying construct. A system that fails to recognize this equivalence produces a false split, missing the connection.

These are not edge cases. They are the normal state of cross-domain scientific knowledge. The problem is that current integration approaches — whether based on long-context synthesis, retrieval-augmented generation (RAG), or schema fusion — lack a principled representation of what exactly is being compared. They conflate referent identity, construct identity, measurement/operationalization equivalence, temporal context, modality, and attribution into a single similarity or alignment score.

1.1 The proposal in brief

ORION Paper III proposes a *recoverable global knowledge portrait* built from *source-local scientific projections*. Rather than extracting facts into a canonical knowledge graph, ORION retains each source's claim as a projection bound to its exact provenance, with separately represented coordinates for referent, construct, measurement, context, polarity, modality, and attribution. A proposed mapping between two projections is not a generic similarity score but a typed assertion that may target specific coordinates. When all relevant coordinates are compatible, projections may be GLUEd into a unified chart. When they are not, the correct output is a *typed obstruction* or *plural portrait* — an informative representation of incompatibility.

The key claim is that this architecture measurably reduces false integration — the erroneous merging of claims that should not be merged — while preserving useful synthesis. We test this claim through a frozen, independently annotated gold dataset spanning four disciplines (biomedicine, physics, social science, climate science) and eight case families designed to stress specific failure modes.

1.2 Claims and non-claims

To bound the novelty space, we explicitly absorb the following as established prior work rather than claimed contributions:

- Source-grounded cross-domain structured knowledge extraction (MUSE [5], Discovery Engine [8])
- Multidisciplinary expert scientific-process schemas (SciSchema.org [6])
- Evidence-linked schema induction and fusion (SCOPE/SCION [9])
- Provenance-aware multi-source knowledge integration (Executable Schema Contracts [10])
- Scientific entity/relation extraction and discourse parsing (SciER [22])
- Ontology and schema alignment [7]
- Entity resolution, data integration, and record linkage [13, 12]
- Contradiction and stance detection [15]
- Literature-based discovery [19]
- Scientific pluralism in philosophy of science [11, 3, 4]
- LLM-based structured knowledge representation [17, 20, 21, 18]

The surviving residual is the claim that *typed coordinate distinctions, conditional GLUE with obstruction, recoverability, and the W effect jointly produce a measurable reduction in false integration* compared with the strongest flat integration baseline. This is a narrow, falsifiable claim that the gold study is designed to test.

1.3 Structure of the paper

Section 2 situates the proposal within the absorbed literature. Section 3 provides formal definitions of the key concepts. Section 4 describes the gold dataset and annotation protocol. Section 5 specifies the evaluation design, baselines, ablations, and hypotheses. Section 7 reports results (currently `CANNOT_CHECK`). Section 9 discusses limitations and reproducibility. Section 11 concludes.

2 Related work

ORION Paper III's contribution is deliberately narrow. Rather than claiming novelty in any single mechanism — extraction, schema matching, data integration, contradiction detection, or literature-based discovery — we situate each as absorbed prior work and identify the precise residual.

2.1 Structured scientific knowledge bases and extraction

MUSE [5] constructs a full-text cross-domain knowledge base of scientific problems, solutions, and rationales, grounding each structure in its originating sentences. Through this mechanism, source-grounded cross-domain structured knowledge extraction is absorbed as prior art. However, MUSE produces a single problem/solution/rationale structure per source; it does not distinguish referent, construct, measurement, context, modality, and attribution as separate typed hypotheses, nor does it represent the failure to align as an obstruction. The Discovery Engine [8] similarly builds structured scientific knowledge artifacts from literature. Both are valuable inputs and baselines for ORION, not competitors to be contested.

SciER [22] provides full-text scientific entity and relation annotations and establishes the extraction quality bar for datasets, methods, and tasks in scientific documents. Scientific information extraction, discourse parsing, coreference resolution, and modality detection are absorbed mechanisms. Its flat entity/relation extraction does not, by itself, represent the participation of typed coordinates in a GLUE-or-obstruction decision; the extraction is a baseline, and the *vertical* separation of semantic coordinates is residual.

2.2 Schema-centric approaches

SciSchema.org [6] provides expert-designed multidisciplinary scientific-process schemas with explicit parameters, measurements, and provenance — absorbing multidisciplinary expert schema design. Its schemas are static per discipline, whereas ORION treats schemas as *source-local, composable, and obstruable* projections assembled per research question (residual D1). Expert-designed schemas do not by themselves guarantee traceability of the path from source to schema (residual D4).

SCOPE and SCION [9] introduce a benchmark and an auditable reference pipeline for evidence-linked schema induction and fusion. Schema induction and conservative fusion are absorbed. SCOPE/SCION fuses schemas to produce a single integrated schema, whereas ORION's GLUE-or-obstruction semantics (residual D3) permit fusion only when all relevant coordinates are compatible; otherwise an obstruction or plural portrait is the correct output. SCOPE/SCION does not represent measurement/operationalization equivalence as a separate coordinate or track non-preserved information.

Executable Schema Contracts [10] demonstrate provenance-aware shared-schema integration and retrieval. Provenance-aware multi-source knowledge integration and retrieval under schema constraints are absorbed. However, Executable Schema Contracts assume a shared schema exists or can be derived; ORION allows the case where no shared schema is justified, producing an obstruction (residual D3), and it extends recoverability beyond provenance to include *non-preserved coordinates* (residual D4).

2.3 Alignment, integration, and resolution

Ontology and schema matching [7] pairs—including LLM-based matching such as LLMatch [21]—find correspondences between schemas or ontologies. The mechanism of finding correspondences is absorbed. Alignment methods typically produce a single equivalence or subsumption relation, whereas ORION's D2 distinguishes referent, construct, measurement, and context as *separate alignment axes*: two schemas may align at the construct level yet differ at the measurement level. Alignment methods do not produce preservation conditions (D4) or obstruction verdicts (D3).

Data integration, entity resolution, and record linkage [13, 12] assume a shared referent and produce a fused record. ORION's D3 admits cases where the correct output is not a fused record, and D2 distinguishes same-referent/different-construct cases that data integration collapses into a single alignment score. Federated and multimodel knowledge graphs [12] treat each graph as canonical for its domain, whereas ORION treats each source as a local, non-canonical projection (D1) and tracks what is lost in any mapping (D4).

2.4 Claim comparison and contradiction

Contradiction and stance detection [15] typically produce a binary contradiction/non-contradiction or stance label. ORION distinguishes contradiction from *context difference*, *modality difference*, *measurement difference*, and *attribution difference* — each of which a standard system might label as "contradiction." Only true contradiction (after all coordinates are aligned) should trigger a revision; the other differences are informative obstructions or condition-aligned mappings.

2.5 Measurement equivalance and construct validity

The psychometric and metrological literatures on measurement equivalence, construct validity, and invariance testing [16] have long established that constructs and their operationalizations are distinct. ORION does not claim to discover this fact; it claims to represent it as a first-class typed coordinate in an automated integration pipeline and to make the GLUE-or-obstruction decision depend on it. The measurement-equivalence literature does not provide a generic computational representation that plugs into a knowledge integration system.

2.6 Literature-based discovery

Swanson's A-B-C bridging [19] and later systems find implicit connections between literatures that do not cite each other. The mechanism of discovering bridges is absorbed. ORION adds the requirement that the meaning of the bridging concept B be checked for compatibility across both halves of the bridge before the composition is accepted: if B changes meaning (referent, construct, measurement, or context) between A-B and B-C, the bridge is an obstruction, not a discovery. This is the residual D2/D3 applied to literature-based discovery.

2.7 Scientific pluralism

The philosophy of science literature on scientific pluralism [11, 3, 4] has argued for decades that multiple incompatible scientific representations can each be valid and that forced unification is harmful. ORION does not claim to discover pluralism; it claims that an automated integration system can *represent* pluralism as a first-class output — OBSTRUCTION or PLURAL_VIEW — rather than being forced to produce a single unified answer. The computational implementation of a pluralism principle is the residual.

2.8 Scientific RAG and cross-domain synthesis

OpenScholar [17] synthesizes scientific literature with retrieval-augmented generation; BioSage [20] performs cross-domain scientific reasoning. Scientific RAG and cross-domain translation are ab-

Mechanism family	**Absorbed (related work)**	**ORION residual**
Source-grounded structured KB	MUSE, Discovery Engine	Source-local addressable projections (D1)
Expert scientific schemas	SciSchema.org	Composable, obstruable projections (D1)
Schema induction/fusion	SCOPE/SCION	Conditional GLUE; obstruction (D3)
Provenance integration	Executable Schema Contracts	Non-preserved coordinates (D4)
Scientific IE	SciER	Typed coordinates gate integration (D2)
Ontology/schema alignment	Ontology matching, LLMatch	Separate alignment axes (D2)
Measurement equivalence	Psychometrics, metrology	First-class computational coordinate (D2)
Data integration	Record linkage, data fusion	Non-merge outputs; recoverability (D3,D4)
Contradiction detection	Stance/contradiction systems	Coordinate-resolved contradiction
Literature-based discovery	Swanson lineage	Meaning-stability check on bridges (D2,D3)
Scientific pluralism	Philosophy of science	Computational representation (D3)
Scientific RAG / cross-domain	OpenScholar, BioSage	Testable additive invariants
FM-based KG representation	Discovery Engine, ADIAS	Architectural invariants vs prompt engineering

Table 1: The absorbed boundary. Each mechanism family is prior art; the residual is the last column.

sorbed mechanisms, and serve as strong baselines in our evaluation. LLM-based structured extraction, in-context integration, and prompt-based schema alignment [18] are likewise absorbed. These systems typically invoke a single prompt or schema to extract and integrate, implicitly flattening source projections into a canonical representation. ORION's architectural invariants — typed coordinates, GLUE-or-obstruction, recoverability — are orthogonal to the prompt engineering question and are testable as additive components on top of any backbone. Whether these invariants measurably reduce false integration relative to flat integration is the empirical question our gold study answers.

2.9 Summary of the absorbed boundary and residual

We do not claim novelty in any row of the middle column. The claimed contribution is that the combination of typed coordinate distinctions, conditional GLUE with typed obstruction and plural portraits, recoverability, and the W effect is not addressed by any single prior mechanism, and that this combination can be evaluated as a measurable reduction in false integration. Table 1 summarizes the boundary.

3 Method

This section defines the ORION Paper III formal objects (ScientificMeaningProjection.v1) precisely enough for reimplementation. The design is registered as protocol `P3.cross-domain-atlas.v1` in `protocol/`. All definitions below are data-level; they constrain what any engine—local exact world, LLM orchestrator, or hybrid—must compute to be evaluated against the protocol.

3.1 Source and corpus

Let $\mathcal{C} = \{d\}$ be a corpus of source documents. Each source document d is identified by a tuple

$$d = \langle \mathrm{id}, v, \mathrm{span}, T, \mathrm{discipline}, h \rangle \tag{1}$$

where id is a document identifier (DOI, PMC ID, or arXiv ID), v is the document version, span = [start : end] locates the relevant passage in the released/authorized text T, discipline is the scientific field, and h is the SHA-256 hash of the span text, fixing the exact bytes a judgment refers to. No judgment is permitted to float free of its $\langle \mathrm{id}, v, \mathrm{span}, h \rangle$ address.

3.2 Source-local scientific projection

For each source $s = \langle d, \mathrm{span}, h \rangle$, ORION computes a *scientific meaning projection*

$$P_s = \langle R_s, C_s, M_s, X_s, \rho_s, \mu_s, a_s, \delta_s, \Pi_s \rangle \tag{2}$$

whose coordinates are:

R_s **(referent):** the entity or class the claim is about;

C_s **(construct):** the theoretical construct invoked;

M_s **(measurement/operationalization):** the instrument, protocol, or operational definition;

X_s **(context):** temporal/state and other context;

ρ_s **(polarity):** positive/negative claim orientation;

μ_s **(modality):** assertion strength or modal force;

a_s **(attribution):** speaker/source attribution;

δ_s **(discourse):** discourse relation relative to the surrounding passage;

Π_s **(assumptions):** explicit assumptions and unresolved ambiguity.

Each coordinate is a *typed hypothesis*, not a hallucinated ground truth. The projection is source-local: it addresses the source's claim in the source's own terms and is not contaminated by other sources or by an a priori global schema. An extraction or normalization step is not permitted to silently acquire scientific authority over the source view.

3.3 Scientific meaning comparison

A comparison between two projections P_a, P_b is a vector of *coordinate-wise hypotheses*:

$$m(P_a, P_b) = \langle \mathrm{ref}(R_a, R_b), \mathrm{const}(C_a, C_b), \mathrm{meas}(M_a, M_b), \mathrm{ctx}(X_a, X_b), \mathrm{pol}(\rho_a, \rho_b), \mathrm{mod}(\mu_a, \mu_b), \mathrm{attr}(a_a, a_b), \mathrm{disc}(\delta_a, \delta_b \tag{3}$$

Each coordinate relation takes values in the frozen taxonomy of `ANNOTATION_SCHEMA_V1.json`, e.g. $\mathrm{ref} \in \{\mathrm{SAME}, \mathrm{DIFFERENT}, \mathrm{PARTIAL\_OVERLAP}, \mathrm{UNRESOLVED}\}$, $\mathrm{const} \in \{\mathrm{SAME}, \mathrm{RELATED\_NOT\_SAM}$ $\mathrm{meas} \in \{\mathrm{EQUIVALENT}, \mathrm{TRANSFORMABLE\_WITH\_CONDITIONS}, \mathrm{NON\_EQUIVALENT}, \mathrm{NOT\_APPLICA}$ A mapping is *proposal-level* until supported.

3.4 Typed mapping

A *typed mapping* is a pair

$$m \in \{\mathrm{IDENTITY}, \mathrm{EQUIVALENCE}, \mathrm{CONDITIONAL\_TRANSFORM}, \mathrm{SUBSUMPTION}, \mathrm{ASSOCIATION\_ONLY} \tag{4}$$

together with

- a set of coordinate relations $\mathrm{comp}(m)$ it licenses (which coordinates it asserts to align);
- a set of *preservation conditions* $\mathrm{pre}(m)$ that must hold for the mapping to be relied upon, e.g. "P_b restricted to temperature range $[T_1, T_2]$"; and
- a set of *non-preserved coordinates* $\overline{\mathrm{pre}}(m)$: information in P_a or P_b that the mapping does not carry, which must not be silently assumed to transfer.

3.5 GLUE and obstruction

A set of projections $\{P_i\}_i$ is *composable* on a mapping set m if every licensed coordinate relation is satisfied and every preservation condition $\mathrm{pre}(m)$ is verified. When composable, the projections may be GLUEd into a joint chart $\bigsqcup_i P_i$ with the mapping record attached. When any licensed relation fails or any preservation condition is violated (and cannot be repaired), the correct output is a *typed obstruction*: the system records OBSTRUCTION with the failing coordinate(s) rather than forcing a merge. When multiple internally valid but mutually incompatible representations persist, the correct output is PLURAL_VIEW. Unresolvable cases take UNRESOLVED; the system is permitted (and encouraged) to abstain rather than guess.

3.6 Recoverable global portrait

A *global portrait* is the structure

$$\mathcal{G} = \langle \{P_i\}, \mathcal{M}, \sigma \rangle \tag{5}$$

where $\{P_i\}$ retains every contributing source projection, $\mathcal{M}$ records every proposed and accepted mapping with its conditions, and σ is an *assignment map* from each global statement back to its source projections, mappings, and surviving deductions. Recoverability is the requirement that for every global statement $g \in \mathcal{G}$ and every source projection P_i, the reader (or automated evaluator) can recover: (a) which source span(s) licensed g; (b) which mappings were applied and under what conditions; and (c) which coordinates were *not* preserved — i.e. what g does *not* assert. Recovery of non-preserved coordinates is an evaluation target, not a display concern.

3.7 The W effect

Let Ω_0 be the relevance/search-universe model of the integration system before absorbing some reader-set S of new sources, and let Ω_1 be the universe model after a projection P_s for $s \in S$ is absorbed. The *W effect* is the claim that $\Omega_1 \supsetneq \Omega_0$: the new representation does not merely add facts to a static graph; it *changes the search universe* — new relevance axes, new addressable referents, new mappings — and thereby reopens hypotheses. Formally, the effect is a feedback loop

$$\Omega_{t+1} = \mathrm{Universe}(\mathcal{G}_t \cup \{P_s : s \in S_t\}) \tag{6}$$

so that integrating new literature can change which future sources are retrieved and how they are posed. Ablation ORION_NO_W_EXPANSION removes this feedback to measure its marginal effect.

Discipline	**Samples**
Biomedicine	8
Physics	8
Social Science	8
Climate Science	8
Total	**32**

Table 2: Discipline distribution of the seed manifest. Balanced 8 per discipline per `case_family_coverage` of `SAMPLE_MANIFEST_SEED_V1.json`.

3.8 What is not claimed

ORION does not claim the first scientific knowledge graph, cross-domain RAG system, schema matcher, literature-based discovery system, or provenance graph. Its target is narrower: whether preserving source-local projections and explicitly representing failed mappings reduces scientifically damaging false integration while still enabling useful synthesis. Local exact worlds [1, 2] falsify the semantics of projection, mapping, and GLUE in controlled settings; they do not by themselves establish adequacy over the real literature, which is why the external gold study is the decisive evaluation (Section 5).

4 Gold dataset

The decisive evaluation of ORION Paper III requires ground truth that challenges an integrator where failure is most damaging. This section documents the *P3.cross-domain-atlas.v1* gold dataset: the corpus design, the eight case families, the annotation schema and handbook, the adjudication policy, and the status of the released seed manifest. The annotation contract is frozen before final gold labeling and before any evaluated system output is inspected (`outcome-blind`; see `protocol/ADJUDICATION_POLICY_V1.md`).

4.1 Corpus design

Sources are selected from open-access publications (PMC, arXiv, or open-access journals) per the licensing rules in `protocol/CORPUS_DESIGN_V1.md`: peer-reviewed, open-access or CC-BY licensed, with stable metadata (DOI, arXiv ID, or PMID) and at least one comparable claim pair per case. The four disciplines were chosen (Table 2) to produce materially different terminology, measurement cultures, and integration failure modes: biomedicine for high-stakes integration and a mature ontology landscape; physics for mathematical precision and multiple interpretation frameworks; social science for construct-measurement diversity and pluralism challenges; climate science for multi-scale integration and attribution complexity. No full-text is redistributed; the release carries document identifiers, exact span addresses, and retrieval instructions, with legally shareable spans where licensing permits.

4.2 Case families

Thirty-two samples are organized into eight case families of four samples each (one per discipline), each designed to stress a distinct failure mode of an integrator:

1. **same_name_different_referent**: two sources use the same surface term for distinct entities (e.g. "zebrafish" as model organism versus invasive species). A naive integrator that merges on surface form false-merges.

2. **different_name_same_referent**: two sources use different surface terms for the same entity or construct. A naive integrator false-splits.

3. **same_construct_different_measurement**: the same underlying construct is measured with different instruments or operationalizations; integration is valid only under explicit transformation conditions.

4. **same_entity_different_temporal_state**: the same entity appears at different time points or states; the apparent discrepancy is *not* a contradiction.

5. **polarity_modality_attribution_context**: claims that share polarity but differ in modality, attribution, or context; resolving them requires the coordinate-wise representation.

6. **valid_invalid_representation_mapping**: a non-isomorphic but valid mapping between two classification or representation systems, where the mapping is legitimate if and only if preservation conditions hold.

7. **valid_invalid_literature_bridge**: an A-B-C bridge between literatures where the transfer is valid only under conditions (Swanson-style), and invalid without them.

8. **genuine_plural_obstruction**: genuinely incompatible scientific representations where no merge should be authorized; the correct verdict is PLURAL_VIEW.

The full seed manifest (`gold/SAMPLE_MANIFEST_SEED_V1.json`) carries 32 structurally valid entries with placeholder document identifiers and hashes. Final gold replaces these with verified open-access spans and full text hashes.

4.3 Annotation schema and handbook

Each sample is annotated on the 11 coordinates of `protocol/ANNOTATION_SCHEMA_V1.json`: `referent_relation`, `construct_relation`, `measurement_relation`, `context_relation`, `polarity_relation`, `modality_relation`, `attribution_relation`, `discourse_relation`, `mapping_relation`, `contradiction_verdict`, and `integration_verdict`. Every coordinate is a typed judgment in a frozen taxonomy, and every annotation binds its verdicts to exact source spans via `source_span` (`document_id`, `document_version`, `span_start`, `span_end`, `text_hash`). The integration verdict is one of `GLUE_ALLOWED`, `OBSTRUCTION`, `PLURAL_VIEW`, or `UNRESOLVED`. `UNRESOLVED` is a first-class, legitimate label: adjudication is not permitted to manufacture certainty.

The written handbook (`protocol/ANNOTATION_HANDBOOK_V1.md`) gives instructions for each coordinate, boundary cases, and the rule that annotators must not inspect evaluated-system outputs. The adjudication sequence (`protocol/ADJUDICATION_POLICY_V1.md`) is: independent double annotation on a shared subset; per-coordinate agreement reporting (not only a single global score); adjudication of disagreements; domain-expert escalation where specialist operationalization knowledge is required; and default to `UNRESOLVED` where agreement cannot be reached. The policy prohibits deleting hard cases, revising the handbook after outcomes are known, and collapsing measurement disagreement into construct disagreement.

4.4 Status and release

The released manifest is `SEED`: structurally valid placeholders awaiting expert annotation. The target states are `GOLD_COMPLETE` (all 32 samples annotated and adjudicated) and `QUALITY_GATE` (inter-annotator agreement at or above threshold on every coordinate). The gold, adjudicated records, and reproducibility checklist will be released under CC-BY 4.0 with the data statement in `gold/README.md`. Reuse of overlapping existing resources (MUSE, SciSchema, SciER) is preferred wherever licenses and task shapes match, per `JOURNAL_READINESS.md`.

5 Evaluation

The evaluation is governed by the frozen protocol `P3.cross-domain-atlas.v1` in `protocol/`. The design, hypotheses, baselines, ablations, metrics, and analysis plan are described below. All systems are evaluated on the same gold dataset (Section 4) across five stochastic seeds. The primary hypothesis was registered before the gold dataset reached `GOLD_COMPLETE` status.

5.1 Primary hypothesis (H1)

The primary hypothesis is a two-part safety and utility claim:

H1a (superiority): $\mathrm{ORION}_{\mathrm{Full}}$ achieves a false merge rate at least $\delta_s = 0.05$ (five percentage points) lower than the strongest matched baseline, tested at $\alpha = 0.05$ with Holm correction over the family of baseline comparisons.

H1b (non-inferiority): $\mathrm{ORION}_{\mathrm{Full}}$ achieves a valid integration rate no worse than $\delta_{ni} = 0.03$ (three percentage points) below the strongest matched baseline, with the same Holm-corrected $\alpha = 0.05$.

Both must pass for the hypothesis to be considered confirmed. The margins $\delta_s = 0.05$ and $\delta_{ni} = 0.03$ are prospective design margins, not claimed results. They are derived from a power analysis targeting 80% power at $\alpha = 0.05$ for a Cohen's $h \geq 0.4$ effect size, inflated by 25% to account for the Holm correction. The margins are not adjusted after seeing the data.

5.2 Secondary hypotheses

H2 (recoverability): ORION achieves a source recoverability rate ≥ 0.80, measured as the proportion of cases where the gold-annotated recoverability target is fully satisfied by the system's output.

H3 (obstruction value): ORION's obstruction precision ≥ 0.70 and obstruction recall ≥ 0.50 on genuine obstruction cases, reflecting the value of typed obstruction as an informative output rather than a failure.

H4 (ablation necessity): For each of the 8 ablations, the direction of the ablation effect (degradation on the relevant metric) is consistent with the coordinate's hypothesized role. Formal hypothesis tests use paired bootstrap difference with Holm correction on the family of ablation comparisons.

5.3 Baselines

Six baselines are implemented in `src/orion/study/baselines.py`:

1. **VanillaLongContext**: a single long-context prompt that presents both source documents and asks for an integration verdict directly, without any intermediate projection or mapping representation. This measures the performance of flat multi-paper synthesis.

2. **ScientificRAG**: a retrieval-augmented generation pipeline that retrieves relevant passages from a corpus of scientific text and synthesizes an answer. Represents the standard RAG baseline.

3. **CrossDomainRAG**: a cross-domain translation/RAG pipeline that maps source claims into a common representation space before integration. Represents cross-domain translation approaches.

4. **FlatUniversalSchema**: a single universal schema is induced from the sources and all claims are projected into it. Represents KG canonicalization and flat universal-schema baselines.

5. **SCOPE_SCION_Like**: an evidence-linked schema induction and fusion baseline, modelled after SCOPE/SCION [9]. Sources are processed through schema induction, and the resulting schemas are fused.

6. **ProvenanceSchema**: a provenance-aware schema-contract integration baseline, modelled after Executable Schema Contracts [10]. Sources are integrated under a shared schema contract with provenance tracking.

All baselines are resource-matched to the extent possible: they use the same underlying model family (deepseek-v4-pro) and the same context and retrieval budgets. The baselines are not straw men; they are competitive approaches from the absorbed prior work.

5.4 Ablations

Eight ablations are implemented in `src/orion/study/ablations.py`, each removing one architectural component from the full ORION system:

1. **ORION_no_referent**: removes the referent identity coordinate from the projection and mapping. Integration decisions proceed without distinguishing referent identity.

2. **ORION_no_construct**: removes the construct identity coordinate. Measures the contribution of the construct dimension.

3. **ORION_no_measurement**: removes the measurement/ operationalization coordinate. Tests whether measurement equivalence is necessary for safe integration.

4. **ORION_no_context**: removes the temporal/state context coordinate. Measures the effect of context awareness.

5. **ORION_no_modality_polarity_attribution_discourse**: removes the modality, polarity, attribution, and discourse coordinates together. Tests the contribution of the discourse-modality complex.

6. **ORION_no_obstruction**: removes the explicit obstruction state; the system must produce a best mapping for every case, even when coordinates do not align. Tests the value of typed obstruction.

7. **ORION_no_recoverability**: removes the recoverability target; the system does not track non-preserved coordinates. Tests the marginal effect of recoverability.

8. **ORION_no_W_expansion**: removes the W expansion feedback loop; the search universe is static. Tests the effect of the W effect.

5.5 Metrics

Seventeen metrics are computed per run (defined in `src/orion/study/metrics.py`):

1. **false_merge_rate**: proportion of OBSTRUCTION cases where the system incorrectly produced GLUE_ALLOWED.

2. **valid_integration_rate**: proportion of GLUE_ALLOWED cases correctly identified.

3. **false_integration_rate_with_valid_integration_guard**: composite of the two primary metrics.

4. **mapping_precision_recall_f1**: per-mapping-type precision, recall, and macro-F1.

5. **source_recoverability_rate**: proportion of cases where the gold recoverability target is satisfied.

6. **preservation_condition_accuracy**: proportion of preservation conditions correctly identified.

7. **measurement_equivalence_accuracy**: accuracy on the measurement/operationalization coordinate.

8. **obstruction_precision**: among cases the system labelled OBSTRUCTION, proportion that are gold OBSTRUCTION.

9. **obstruction_recall**: among gold OBSTRUCTION cases, proportion correctly identified.

10. **unresolved_abstention_calibration**: proportion of UNRESOLVED outputs that match gold UNRESOLVED.

11. **annotation_agreement_by_coordinate**: agreement between the system's per-coordinate judgments and the gold.

12. **cost_metrics**: wall-clock time, token count, storage overhead, and estimated currency cost.

All metrics are reported with Wilson 95% confidence intervals and aggregated across five seeds via bootstrap mean with 95% CI.

5.6 Statistical analysis

The analysis follows the frozen plan in `protocol/PROTOCOL_V1.json`:

1. **Primary test**: paired bootstrap difference (10,000 resamples) between $\text{ORION}_{\text{Full}}$ and the strongest baseline on false merge rate, testing superiority margin $\delta_s = 0.05$. Holm correction across the six baseline comparisons. Non-inferiority test on valid integration rate with margin $\delta_{ni} = 0.03$.

2. **Ablation tests**: for each ablation, paired bootstrap difference against $\text{ORION}_{\text{Full}}$ on the relevant metric, with Holm correction over the 9 comparisons. Effect sizes reported as Cohen's h with categories (negligible/≤ 0.2, small/≤ 0.5, medium/≤ 0.8, large/> 0.8).

3. **Discipline-stratified**: the primary metrics are computed separately per discipline. A discipline-by-discipline breakdown is reported with the pooled result.

4. **Safety margin**: $\text{ORION}_{\text{Full}}$ must pass *both* the superiority and non-inferiority tests at the same $\alpha = 0.05$ for H1 to be confirmed.

5. **Secondary analyses**: secondary hypotheses (H2–H4) are tested at $\alpha = 0.05$ without correction, with effect sizes reported for interpretation.

5.7 Reproducibility

All evaluation outputs are emitted as normalized result records (schema: `orion.publication-result-record.v1`) with a run manifest (schema: `orion.run-manifest.v1`) that records the gold hash, system versions, seeds, and resource policy. The MUSE reference implementation is pinned to commit `f7a40317db46145d0c90b221311d8324db5da1b9` [5]. The evaluation runner, `src/orion/study/evaluate.py`, produces a deterministic output given the same gold, system versions, and seeds. The reproducibility package includes the annotation handbook, gold with provenance, baseline configurations, raw projections, and scripts for every metric, figure, and table.

6 Results

7 Results status

The ORION Paper III implementation is currently at `CANNOT_CHECK` status for the primary empirical claims. This section describes what has been verified, what remains unverified, and the implementation gaps that must be addressed before the gold study can be executed.

7.1 Verified: local exact-world semantics

The ORION Projection Ontology v1 has been implemented and tested against synthetic local exact-world scenarios. These tests verify:

1. Correct projection semantics for referent identity (same vs. different referent), construct identity, measurement equivalence, and context differences.

2. Correct GLUE gate logic: GLUE is permitted only when all compatibility conditions are satisfied.

3. Correct obstruction classification: genuine incompatibility produces OBSTRUCTION; valid alternative perspectives produce PLURAL_VIEW.

4. Correct mapping hypothesis generation: typed mapping hypotheses are produced for each of the 11 coordinates.

5. Correct preservation condition tracking: transformed and non-preserved coordinates are enumerated.

These tests use synthetic source projections that are pre-structured in the ORION format. They demonstrate that the projection ontology and mapping calculus are implementable and produce correct semantics for the targeted cases. They do not establish that the framework works with real scientific text.

7.2 Identified gap: text-to-scientific-meaning layer

The local exact-world tests exposed a critical missing component: a *text-to-scientific-meaning extraction layer* that converts natural language scientific text into the structured 11-coordinate projection format. This layer is not part of the current ORION codebase.

The extraction task is nontrivial because it requires:

1. Identifying the referent entity or phenomenon from the source text.
2. Distinguishing the theoretical construct from its operationalisation.
3. Extracting the measurement procedure, threshold, and units.
4. Determining the temporal, spatial, and experimental context.
5. Classifying the polarity, modality, attribution, and discourse role of each claim.
6. Resolving these distinctions across sentences and paragraphs, not just within a single sentence.

Without this layer, the ORION framework cannot process real scientific papers, and the end-to-end evaluation against baselines cannot be performed. The extraction layer is listed as a reproducibility prerequisite.

7.3 Remaining unverified claims

The following claims remain `CANNOT_CHECK` until the gold study is executed under the bound V1 protocol:

- **False-merge reduction (H1):** Whether ORION's typed coordinates reduce false integration relative to flat baselines.
- **Recoverability (H2):** Whether the recoverability target measurably improves the reader's ability to trace source provenance.
- **Obstruction value (H3):** Whether explicitly retaining incompatible charts improves downstream scientific answering.
- **Coordinate necessity (H4):** Whether removing individual coordinates causes measurable errors.

7.4 Downstream benefit

The intended downstream benefit of the ORION framework is improved scientific decision-making: fewer false merges mean fewer incorrect conclusions drawn from integrating incompatible sources; explicit obstruction means unresolved disagreements are recorded rather than hidden; recoverability means the reasoning behind a global statement can be audited. These benefits have not been measured yet and remain the target of the gold study.

7.5 Needs-improvement analysis

The current implementation has three specific needs-improvement items:

1. **Missing extraction layer:** As noted above, text-to-scientific-meaning extraction is unimplemented. This is the highest-priority gap.
2. **Missing recoverability implementation:** The provenance store, mapping log, and coordinate tracing function are not yet implemented. The recoverability target is defined but not operationalised.
3. **Missing W-effect implementation:** The W-expansion/reopen mechanism is specified but not implemented. The ablation test for D5 cannot be run.

These gaps are not fundamental limitations of the framework; they are engineering tasks that must be completed before the gold study. The framework's formal semantics are verified; the pipeline from text to projection to integration is incomplete.

8 Limitations, ethics and reproducibility

9 Limitations, ethics, and reproducibility

9.1 Limitations

Expert subjectivity and theory-ladenness. Scientific identity and measurement equivalence judgments are theory-laden and may depend on the annotator's disciplinary background. Two experts may reasonably disagree on whether a construct is "the same" across two sources, particularly when the sources operationalise the construct differently. The annotation protocol mitigates this through a written handbook, independent double annotation, and domain-expert escalation, but cannot eliminate subjectivity entirely. Inter-annotator agreement per coordinate will be reported, and the UNRESOLVED label is available for cases where agreement cannot be reached.

Domain coverage. The gold dataset covers four disciplines (biomedicine, physics, social science, climate science) with 32 samples. This is sufficient for the primary hypothesis test but cannot establish universal cross-domain adequacy. The eight case families are designed to span the most challenging integration scenarios, but other scenarios (e.g., cross-linguistic terminology differences, historical conceptual change) are not covered.

Evolving terminology. Scientific terminology evolves over time. A construct that was operationalised one way in 2010 may be operationalised differently in 2026. The annotation captures the state of the literature at the time of annotation; the framework's ability to handle temporal drift in terminology is not evaluated.

Ontology and schema dependence. The 11-coordinate annotation schema is a choice. Other coordinate systems are possible (e.g., adding probability distributions, causal structure, or ethical considerations). The evaluation does not claim that the 11-coordinate schema is optimal — only that it is a principled starting point for typed scientific integration.

Extraction error propagation. The text-to-scientific-meaning extraction layer is the highest-risk component of the pipeline. Errors in this layer propagate to every downstream decision (mapping, GLUE, obstruction). The gold study evaluates the full pipeline end-to-end, so extraction errors will be reflected in the headline metrics; attribution to individual stages will require the per-coordinate metrics and qualitative error analysis.

Limited simultaneous coherence. The ultimate purpose of a global knowledge portrait is to enhance the capacity for scientific research — to expose the epistemic structure of a field, its incompatibilities, and its open questions. The gold study measures recoverability, obstruction quality, and downstream answer correctness, but no study of realistic size can capture the full cognitive and collaborative benefit of such a portrait.

9.2 Ethics

The ORION framework integrates scientific claims from published literature; it does not collect new human data. The gold dataset draws on published open-access sources, so no additional consent or ethics review is required. The annotation task involves expert scientists evaluating scientific claims; it does not involve sensitive personal data. We nonetheless note two ethical considerations. First, the framework's outputs — including OBSTRUCTION and PLURAL_VIEW verdicts — should be presented to users as structured records of incompatibility, not as judgments that one perspective is superior. Second, the provenance mechanism is designed to make every claim attributable to its source, supporting scientific accountability.

9.3 Reproducibility

The evaluation protocol is frozen in `protocol/PROTOCOL_V1.json`, and the annotation schema in `protocol/ANNOTATION_SCHEMA_V1.json`. The annotation handbook and adjudication policy are frozen as `ANNOTATION_HANDBOOK_V1.md` and `ADJUDICATION_POLICY_V1.md` before any test-system outputs are examined. The MUSE reference implementation is pinned to a specific commit. Final sampled-source, gold, evaluator, model, and subject hashes remain unbound until `EXECUTION_FROZEN`. The full reproducibility requirements are documented in the repository's reproducibility sections of the paper and its companion materials [14].

10 Conclusion

11 Conclusion

ORION Paper III asks whether a global knowledge portrait — an integration of scientific claims across disciplines and measurement systems — can be built safely. The framework's answer is that safe integration requires three structural commitments that prior approaches lack.

First, *projection before mapping.* Each source retains a local projection whose scientific meaning is recorded in 11 typed coordinates, bound to exact provenance, before any cross-source decision is made. This keeps the question of what a source means separate from the question of which sources can be combined.

Second, *GLUE-or-obstruction*. Integration is a forced decision over typed mapping hypotheses: projections are merged only when all coordinate conditions are satisfied, and otherwise the system reports a typed obstruction or retains a plural portrait. The output of the framework is informative even when no single answer exists.

Third, *recoverability*. The global portrait must expose its own construction — source projections, mapping path, preservation conditions, and non-preserved coordinates — so that a downstream consumer can audit any claim back to its origins.

The formal semantics of the projection-mapping-GLUE calculus have been verified in local exact worlds. What remains is the engineering: a text-to-scientific-meaning extraction layer, a recoverability store, and a W-expansion mechanism, followed by the frozen cross-domain gold study. The evaluation protocol registered here commits us to a specific, falsifiable claim: that ORION reduces false integration relative to strong baselines while preserving valid integration, at acceptable cost.

If the gold study confirms the hypothesis, the ORION framework offers a general response to the central difficulty of scientific knowledge integration — that legitimate differences between communities must be represented, not flattened. If it does not, the framework's formal apparatus will have failed an empirical test that the literature on knowledge integration has rarely, if ever, subjected its methods to. Either outcome advances the design space.

The defences that usually excuse forced unification — that a universal schema is tractable, that a similarity score is sufficient, that integration is always better than obstruction — are exactly the assumptions this work takes to be unverified. They are hypotheses about scientific discourse, and they deserve to be treated as such.

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